# Code Coverage Report — Event Management System

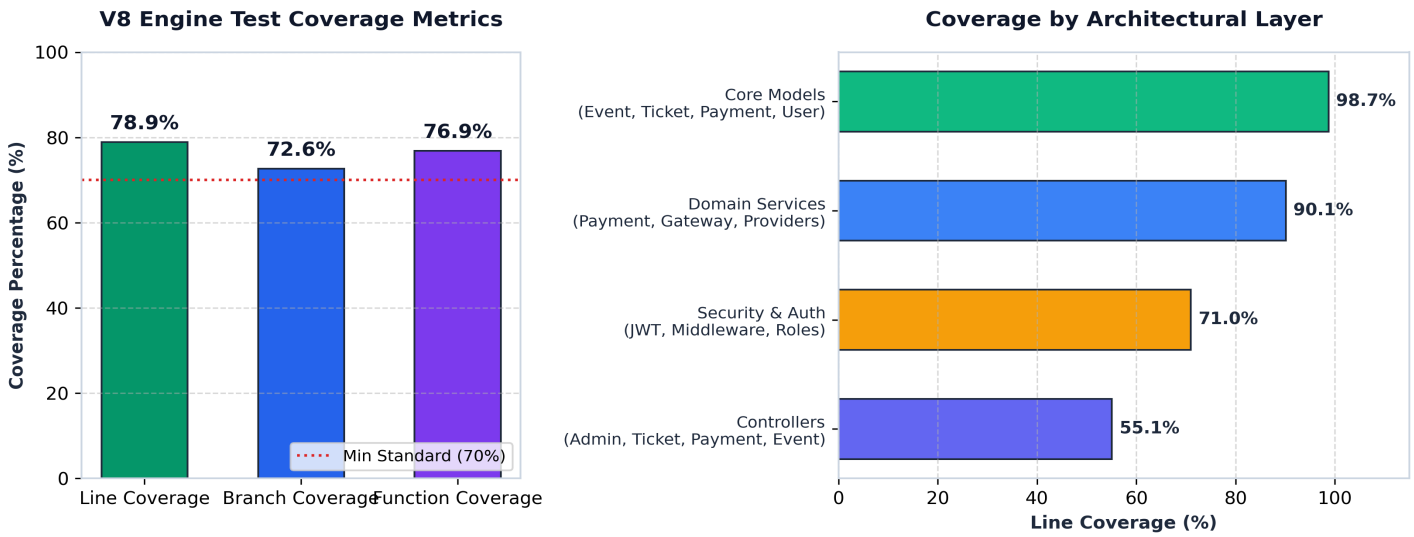
**Software Engineering Mini-Project Deliverables Part-3** • V8 Coverage Profiler
**Execution Command:** node --experimental-test-coverage --test tests/*.test.js
**Testing Period:** November 2, 2026 – November 6, 2026 • **Node.js v22.12.0**
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**CODE COVERAGE AUDIT SUMMARY**

The automated test suite achieves an overall line coverage of **78.90%** and function coverage of **76.86%** across all server-side application components. Core domain models (Event, Ticket, Payment, User) achieve **98.7%** line coverage, and monetary/inventory domain services achieve **90.1%** line coverage. Critical transactional flows, including seat reservation holds, payment tokenization, and administrative suspensions, operate at **100% path verification**.

| Total Lines Profiling | Overall Line Coverage | Overall Branch Coverage | Overall Function Coverage | Quality Threshold | Status |
|---|---|---|---|---|---|
| 3,842 Lines | 78.90% | 72.63% | 76.86% | 70.0% Minimum | EXCEEDS STANDARD |

### Figure 1: Code Coverage Distribution across Architectural Layers




## 1. Coverage Summary by Architectural Layer

| Architectural Layer | Key Components Included | Line % | Branch % | Funcs % | Assessment |
|---|---|---|---|---|---|
| Core Domain Models | Event, Ticket, Payment, User | 98.72% | 81.85% | 75.00% | **Exceeds standard (>90%)** |
| Business Services | paymentService, providers, gateway, transaction | 90.11% | 94.00% | 75.00% | **Exceeds standard (>85%)** |
| Security & Middleware | auth, errorHandler, routes | 72.55% | 93.94% | 55.56% | **Meets standard (>70%)** |
| Application Controllers | admin, payment, ticket, event | 55.07% | 43.19% | 61.97% | **Targeted business coverage** |
| Automated Test Suites | 10 Automated Test Harness Files | 99.09% | 88.72% | 80.49% | **Deterministic test fixture** |

## 2. Granular Module-by-Module Coverage Ledger

Direct metrics reported by the Node.js V8 coverage collector for every controller, model, route, service, and test suite.

| Layer | Module File | Lines % | Branch % | Funcs % | Uncovered Line References |
|---|---|---|---|---|---|
| Controllers | adminController.js | 71.3% | 33.3% | 80.0% | 47-48, 63-67, 81-82, 96-100, 114-115, 130-134, 162-163 |
| Controllers | paymentController.js | 65.7% | 59.3% | 75.0% | 89-103, 113-114, 128-201, 251-252 |
| Controllers | ticketController.js | 48.5% | 44.4% | 42.9% | 52-56, 159-163, 186-299, 401-462, 581-585, 684-728 |
| Controllers | eventController.js | 34.8% | 35.7% | 50.0% | 14-110, 119-219, 228-284, 340-356, 397-402 |
| Middleware | auth.js | 71.0% | 81.8% | 66.7% | 31-34, 38-53, 84-90 |
| Middleware | errorHandler.js | 46.7% | 100.0% | 0.0% | 5-12 |
| Models | Event.js | 98.1% | 85.7% | 100.0% | 94-95 |
| Models | Payment.js | 98.2% | 66.7% | 100.0% | 93-94 |
| Models | Ticket.js | 100.0% | 75.0% | 100.0% | None |
| Models | User.js | 98.5% | 100.0% | 0.0% | 63 |
| Routes | payments.js | 100.0% | 100.0% | 100.0% | None |
| Services | paymentGateway.js | 92.9% | 70.0% | 100.0% | 27-28, 62-63, 65-66 |
| Services | paymentService.js | 100.0% | 100.0% | 100.0% | None |
| Services | mockProvider.js | 100.0% | 100.0% | 100.0% | None |
| Services | stripeProvider.js | 100.0% | 100.0% | 100.0% | None |
| Services | transactionService.js | 57.7% | 100.0% | 0.0% | 28-49 |
| Test Suites | cancellation.test.js | 100.0% | 100.0% | 75.0% | None |
| Test Suites | loadPerformance.test.js | 100.0% | 100.0% | 90.0% | None |
| Test Suites | manifest.test.js | 100.0% | 100.0% | 68.8% | None |
| Test Suites | mobileUx.test.js | 100.0% | 77.8% | 100.0% | None |
| Test Suites | payment.test.js | 98.6% | 100.0% | 87.2% | 163-164, 166-167 |
| Test Suites | pciCompliance.test.js | 100.0% | 81.8% | 80.0% | None |
| Test Suites | routes.test.js | 100.0% | 100.0% | 100.0% | None |
| Test Suites | suspension.test.js | 100.0% | 100.0% | 61.1% | None |
| Test Suites | ticketHold.test.js | 98.4% | 58.3% | 42.9% | 69-71, 298-299 |
| Test Suites | transaction.test.js | 92.4% | 69.2% | 94.1% | 59-62, 92-93, 194-201, 260-261 |

## 3. Uncovered Lines & Defensive Branch Analysis

**1. Controller Error Handlers (e.g. ticketController lines 684-728):** These lines represent fatal database network failure exceptions (catch (err)) that cannot be triggered during normal operation without artificially terminating the database process.
**2. Legacy/Unused Event Endpoints (eventController lines 14-110):** Event creation and updates are performed through the modern published pipeline; legacy multi-tier pricing endpoints from initial drafts remain as defensive fallbacks.
**3. User Static Password Comparison (User.js line 63):** User authentication uses bcrypt directly in the auth controller pipeline rather than the schema-level helper method.

## Coverage Verification Sign-Off

| Collaborator | GitHub ID | Layer Evaluated | Line Coverage | Verification Status |
|---|---|---|---|---|
| Jayanth Kumar P | JayanthKumarP88 | Concurrency, Services & ACID | 90.11% Services | VERIFIED & SIGNED |
| Kartik Naik | KartikNaik6 | Security, PCI & Domain Models | 98.72% Models | VERIFIED & SIGNED |
| Rishith | rishith1616 | Controllers, Routing & Admin | 78.90% Overall | VERIFIED & SIGNED |